

Fluconazole 150mg Capsule

DESCRIPTION AND COMPOSITION:

Light blue / Light blue colored size “1” hard gelatin capsule and containing white powder.

Each capsule contains: Fluconazole 150 mg

ACTIONS:

Fluconazole is a member of a new class of triazole antifungal agents. It is a potent and specific inhibitor of fungal sterol synthesis.

Fluconazole is a highly selective inhibitor of fungal cytochrome P-450 sterol C-14 alphas-demethylation. Mammalian cell demethylation is much less sensitive to fluconazole inhibition. The subsequent loss of normal sterols correlates with the accumulation of 14 alpha-methyl sterols in fungi and may be responsible for the fungistatic activity of fluconazole. It has been shown that fluconazole 50mg daily given up to 28 days does not affect testosterone plasma concentrations in males or steroids concentrations in females of childbearing age.

Fluconazole 200 – 400 mg daily has no clinically significant effect on endogenous steroid levels or on ACTH-stimulated response in healthy male volunteers. Interaction studies with antipyrine indicate that single or multiple doses of fluconazole 50 mg do not affect its metabolism.

PHARMACOLOGY:

The pharmacokinetic properties of fluconazole are similar following administration by the intravenous or oral routes. Fluconazole is well absorbed after oral administration. The bioavailability of orally administered fluconazole is more than 90% compared with intravenous administration. Oral absorption is not affected by concomitant food intake. In the fasting state, peak plasma concentrations occur between 0.5 and 1.5 hours post-dose with a plasma elimination half-life of approximately 30 hours. Fluconazole plasma concentrations are dose proportional.

Ninety percent steady-state levels are reached by day 4-5 with multiple once-daily dosing. Administration of a loading dose (on day 1) of twice the usual daily dose results in plasma concentration close to steady state by day 2. The apparent volume of distribution of fluconazole approximates to total body water. Plasma protein binding is low (11-12%).

Fluconazole shows good penetration into all body fluids studied. The concentrations of fluconazole in saliva and sputum are equal to plasma concentrations. In patients with fungal meningitis, fluconazole concentrations in the CSF are approximately 80% of the corresponding plasma concentrations.

The primary route of excretion is renal, with approximately 80% of the administered

dose appearing in the urine as unchanged drug. About 11% of the dose is excreted in the urine as metabolites. Fluconazole clearance is proportional to creatinine clearance. There is no evidence of circulating metabolites.

The long plasma elimination half-life provides the basis for single-dose therapy for vaginal candidiasis and once-daily dosing for all other indications.

Microbiology: Fluconazole administered orally and intravenously was active in variety of animal models of fungal infection. Activity has been demonstrated against opportunistic mycoses, eg infections with *Candida* sp, including systemic candidiasis in immunocompromised animals; with *Cryptococcus neoformans*, including intracranial infections; with *Microsporum* sp; and with *Trichophyton* sp. Fluconazole has also been shown to be active in animal models of endemic mycoses, including infections with *Blastomyces dermatitides*; with *Coccidioides immitis*; including intracranial infection; and with *Histoplasma capsulatum* in normal and immunosuppressed animals.

INDICATIONS:

Fluconazole is indicated for the treatment of the following condition:-

1. Cryptococcosis, including cryptococcal meningitis and infections of other sites (eg. pulmonary, cutaneous). Normal hosts and patients with AIDS, organ transplants or other causes of immunosuppression may be treated. Fluconazole can be used as maintenance therapy to prevent relapse of cryptococcal disease in patients with AIDS.
2. Systemic candidiasis including candidemia, disseminated candidiasis and other invasive candidal infections. These include infections of peritoneum, endocardium and pulmonary and urinary tracts. Candidal infections in patients with malignancy, in intensive care units or those receiving cytotoxic or immunosuppressive therapy may be treated.
3. Mucosal candidiasis including oropharyngeal, esophageal, non-invasive bronchopulmonary infections, candiduria, mucocutaneous and chronic oral atrophic candidiasis (denture sore mouth). Normal hosts and patients with compromised immune function may be treated.
4. Vaginal candidiasis, acute or recurrent.
5. Prevention of fungal infections in patients with malignancy who are predisposed to such infections as a result of cytotoxic chemotherapy or radiotherapy.
6. Dermatomycosis including tinea pedis, tinea corporis, tinea cruris, pityriasis versicolor and candida infections.

CONTRAINDICATIONS:

Contraindicated in patients who have shown hypersensitivity to fluconazole or to related azole compounds or any other ingredient in formulation.

Co-administration of terfenadine or cisapride is contraindicated in patients receiving fluconazole.

PRECAUTIONS/WARNINGS:

1. Fluconazole has been associated with rare cases of serious hepatic toxicity including fatalities, primarily in patients with serious underlying medical conditions. In cases of fluconazole-associated hepatotoxicity, no obvious relationship to total daily dose, duration of therapy, sex or age of patients has been observed. Fluconazole hepatotoxicity has usually been reversible on discontinuation of therapy. Patients who develop abnormal liver function tests during fluconazole therapy should be monitored for the development of more serious hepatic injury. If clinical signs or symptoms consistent with liver disease develop, fluconazole should be discontinued.
2. Fluconazole should be administered with caution to patients with these potentially proarrhythmic conditions.
3. Patients have rarely developed exfoliative cutaneous reactions eg, Steven-Johnson syndrome and toxic epidermal necrolysis, during treatment with fluconazole. AIDS patients are more prone to the development of severe cutaneous reactions to many drugs. If a rash, which is considered attributable to fluconazole, develops in a patient treated for a superficial fungal infection, further therapy with this agent should be discontinued. If patients with invasive /systemic fungal infections develop rashes, they should be monitored closely and fluconazole discontinued if bullous lesions or erythema multiforme develop. The co-administration of fluconazole at doses <400mg/day with terfenadine should be carefully monitored. In rare cases, anaphylaxis has been reported.
4. Some azoles, including fluconazole, have been associated with prolongation of the QT interval of the electrocardiograms. There have been very rare cases of QT prolongation and torsades de pointes in patients taking fluconazole. This is maybe caused by seriously ill patients with multiple confounding risk factors eg, structural heart disease, electrolyte abnormalities and concomitant medications.

PREGNANCY AND LACTATION:

Use During Pregnancy

There have been reports of spontaneous abortion and congenital abnormalities in infants whose mothers were treated with 150mg of fluconazole as a single or repeated dose in the first trimester.

Use in pregnancy should be avoided except in patients with severe or potentially life-threatening fungal infections in whom Fluconazole 150 capsule may be used if the anticipated benefit outweighs the possible risk to the fetus. If this drug is used during pregnancy, or if the patient becomes pregnant while taking the drug, the patient should be informed of the potential hazard to the fetus.

Effective contraceptive measures should be considered in women of child-bearing potential and should continue throughout the treatment period and for approximately 1 week (5 to 6 half-lives) after the final dose.

There have been reports of multiple congenital abnormalities in infants whose mothers were treated with high-dose (400mg/day to 800mg/day) fluconazole therapy for coccidioidomycosis (an unapproved indication). The relationship between fluconazole use and these events is unclear. Adverse fetal effects have been seen in animals only at high-dose levels associated with maternal toxicity. There were no fetal effects at 5 mg/kg or 10 mg/kg; increases in fetal anatomical variants (supernumerary ribs, renal pelvis dilation) and delays in ossification were observed at 25 mg/kg and 50 mg/kg and higher doses. At doses ranging from 80 mg/kg (approximately 20-60 times the recommended human dose) to 320 mg/kg, embryoletality in rats were increased and fetal abnormalities included wavy ribs, cleft palate and abnormal craniofacial ossification.

Case reports describe a distinctive and a rare pattern of birth defects among infants whose mothers received high dose (400-800mg/day) fluconazole during most or all of the first trimester of pregnancy. The features seen in these infants include brachycephaly, abnormal facies, abnormal calvarial development, cleft palate, femoral bowing, thin ribs and long bones, arthrogryposis, and congenital heart disease.

Use During Lactation

Fluconazole is found in human breast milk at concentrations similar to plasma. Breast-feeding may be maintained after a single dose of 150mg fluconazole. Breast-feeding is not recommended after repeated use or after high-dose fluconazole

Effects on ability to drive and use machines:

No adverse effects on the ability to drive or operate machines have been reported.

SIDE EFFECTS:

Fluconazole is generally well tolerated. The undesirable effects associated with fluconazole are: Nervous System Disorders: Headache, dizziness, seizures, taste perversion

Skin and Subcutaneous Tissue Disorders: Rash, alopecia, exfoliative skin disorders including Stevens-Johnson syndrome and toxic epidermal necrosis.

Gastrointestinal Disorders: Abdominal pain, diarrhea, flatulence, nausea, dyspepsia, vomiting. In some patients, particularly those with serious underlying disease such as AIDS and cancer, changes in renal and haematological function test results and hepatic abnormalities have been observed during treatment with fluconazole and comparative agents, but the clinical significance and relationship to treatment is uncertain.

Hepatobiliary Disorders: Hepatic toxicity including rare cases of fatalities, elevated alkaline phosphatase, elevated bilirubin, elevated SGOT, elevated SGPT, hepatic failure, hepatitis, hepatocellular necrosis, jaundice.

Blood and Lymphatic System Disorders: Leukopenia including neutropenia and agranulocytosis, thrombocytopenia.

Immune system Disorders: allergic reaction: anaphylaxis (including angioedema, face oedema, pruritus), urticaria.

Metabolism and Nutrition Disorders: Hypercholesterolaemia, hypertriglyceridaemia, hypokalaemia.

Cardiac Disorders: QT prolongation, torsade de pointes.

DRUG INTERACTIONS:

Coumarin-type anticoagulants

In an interaction study, fluconazole increased the prothrombin time (12%) following warfarin administration in healthy male volunteers. Therefore, careful monitoring of prothrombin time in patients receiving fluconazole and coumarin-type anticoagulants is recommended.

Oral Sulfonylureas

There have been reports that an interaction exists when fluconazole is administered concomitantly with oral sulfonylureas (chlorpropamide, glibenclamide, glipizide and tolbutamide), leading to prolonged serum half-life. Fluconazole and oral sulfonylureas may be co-administered to diabetic patients, but the possibility of a hypoglycaemic episode should be borne in mind.

Hydrochlorothiazide

In a kinetic interaction study, concomitant administration of fluconazole and hydrochlorothiazide increased plasma concentrations of fluconazole by 40%. It is not necessary to change the fluconazole dose regimen in subjects receiving concomitant diuretics, although the prescriber should bear it in mind.

Phenytoin

Fluconazole increases the plasma concentrations of phenytoin. If it is necessary to administer both drugs concomitantly, phenytoin concentrations should be monitored and the phenytoin dose adjusted to maintain therapeutic levels.

Oral Contraceptives

Multiple dose use of fluconazole is unlikely to have an effect on the efficacy of the combined oral contraceptive.

Rifampicin

Rifampicin enhances the metabolism of concurrently administered fluconazole (25% decrease in the AUC and 20% shorter half-life of fluconazole). An increase of the

fluconazole dose should be considered in patients receiving concomitant rifampicin.

Cyclosporin

Fluconazole may significantly increase cyclosporin levels in renal transplant patients with or without renal impairment. Careful monitoring of cyclosporin concentrations and serum creatinine is recommended in patients receiving fluconazole and cyclosporin.

Theophylline

Fluconazole may decrease the clearance rate of theophylline.

Rifabutin

Concomitant administration of fluconazole and rifabutin may increase the serum levels of rifabutin. Careful monitoring of rifabutin serum levels is recommended in patients taking fluconazole and rifabutin.

Tacrolimus

Concomitant administration of fluconazole and tacrolimus may increase the serum levels of tacrolimus. There have been reports of nephrotoxicity in patients to whom fluconazole and tacrolimus were co-administered. Careful monitoring of tacrolimus serum concentrations is recommended in patients receiving fluconazole and tacrolimus concomitantly.

Zidovudine

Concomitant administration with fluconazole shows increased levels of zidovudine most likely caused by the decreased conversion of zidovudine to its major metabolite. Patients receiving fluconazole and zidovudine concomitantly should be carefully monitored for the development of zidovudine-related adverse reactions.

Benzodiazepines

(Short acting) After oral administration of midazolam, fluconazole resulted in substantial increases in midazolam levels and psychomotor effects. Consideration should be given to decreasing the benzodiazepine dosage and the patients should be appropriately monitored if concomitant benzodiazepine therapy is necessary in patients receiving fluconazole.

Azithromycin

There is no significant pharmacokinetic interaction between fluconazole and azithromycin.

Cisapride

There have been reports of cardiac events, including torsades de pointes, in patients to whom fluconazole and cisapride were co-administered. Cisapride increases the risk of ventricular arrhythmia troubles, notably torsades de pointes. A controlled study found that concomitant fluconazole 200mg once daily and cisapride 20 mg 4 times a day yielded a significant increase in cisapride plasma levels and prolongation of QTc interval. Co-administration of cisapride is contraindicated in patients receiving fluconazole.

Pimozide

Combination with fluconazole will increase risk of ventricular arrhythmia troubles, notably torsades de pointes.

Terfenadine

Serious cardiac dysrhythmias secondary to prolongation of QTc intervals in patients receiving azole antifungals in conjunction with terfenadine. The combined use of fluconazole at doses ³ 400mg with terfenadine is contraindicated. The Coadministration of fluconazole at doses <400mg/day with terfenadine should be carefully monitored.

Others

There have been reports that when oral fluconazole is co-administered with food, cimetidine, antacids or following total body irradiation for bone marrow transplantation, no clinically significant impairment of fluconazole absorption occurs.

Concomitant administration of fluconazole and astemizole or other drugs metabolised by the cytochrome P450 system may be associated with elevations in serum concentrations of these drugs. Therefore, caution should be used when co-administering fluconazole.

DOSAGE AND ADMINISTRATION:

Adults:

Cryptococcal Meningitis and Cryptococcal Infections at Other Sites: 400 mg on the first day followed by 200-400 mg daily usually for at least 6-8 weeks.

Prevention of Relapse of Cryptococcal Meningitis in Patients with AIDS: Fluconazole may be administered indefinitely at a daily dose of at least 200 mg after the patient receives a full course of primary therapy.

Candidaemia, Disseminated Candidiasis and Other Invasive Candidal Infections: 400 mg on the first day followed by 200 mg daily. The dose may be increased to 400 mg daily depending on the clinical response. Duration of treatment depends on the clinical response.

Oropharyngeal Candidiasis: 50-100 mg once daily, given for 7 – 14 days. If necessary, treatment should not be normally exceed 14 days except in severely immunocompromised patients.

Atrophic Oral Candidiasis Associated with Dentures: 50 mg once daily for 14 days administered concurrently with local antiseptic measures to the denture.

Other Candidal Infections of The Mucosa, (except vaginal candidiasis), eg oesophagitis, noninvasive bronchopulmonary infections, candiduria, mucocutaneous candidiasis, etc.: 50 mg daily for 14-30 days. The dose may be increased to 100 mg dose in unusually difficult cases of mucosal candidal infections.

Vaginal Candidiasis: 150 mg should be administered as a single oral dose. To reduce the incidence of recurrent vaginal candidiasis, a 150 mg once monthly dose may be used, the duration of therapy should be individualized, but ranges from 4-12 months. Some patients may require a more frequent dosing.

For the prevention of candidiasis, the recommended dosage is 50 – 400 mg once daily. It is based on the patient's risk of developing fungal infection.

For patients at high risk of systemic infection, eg patients who are anticipated to have profound or prolonged neutropenia, the recommended dosage is 400 mg once daily. Fluconazole administration should start several days before the anticipated onset of neutropenia, and continuing for 7 days after the neutrophil count rises above 1000 cells/mm³.

The recommended dosage is 150 mg once weekly or 50 mg once daily for dermal infections including corporis, tinea pedis, cruris and candida infections. Duration of treatment is usually 2-4 weeks but tinea pedis may require treatment for up to 6 weeks.

Tinea versicolor: Recommended dose is 300mg once weekly for 2 weeks; a 3rd weekly dose of 300mg may be needed in some patients, whereas, in some patients, a single dose of 300-400 mg may be sufficient. An alternate dosing regimen is 50 mg once daily for 2-4 weeks.

Elderly

If there is no evidence of renal impairment, normal recommended doses should be used. For patients with renal impairment (creatinine clearance < 40 ml/min) the dosage schedule should be adjusted as follows:

Patient with Renal Impairment: Fluconazole is cleared primarily by renal excretion

as unchanged drug.

There is no need to adjust single dose therapy. In multiple-dose therapy, normal doses should be given on days 1 and 2 of treatment, and thereafter the dosage intervals of daily dose should be modified in accordance with creatinine clearance as follows:

Creatinine Clearance (mL/min)	Dosage interval / daily dose
> 50	24 hours (normal dosage regimen)
< 50	48 hours or 1/2 normal daily dose
Patients receiving regular dialysis	1 normal dose after every dialysis session

ROUTE OF ADMINISTRATION:

Oral administration

SYMPTOMS AND TREATMENT FOR OVERDOSAGE:

In the event of overdosage, symptomatic treatment (with supportive measures and gastric lavage if clinically indicated) should be instituted. Forced diuresis would probably increase the elimination rate as fluconazole is largely excreted in urine. A three-hour haemodialysis session decreases plasma levels by approximately 50%.

PRESENTATION:

Blisters of 1 capsule. Boxes of 1 and 20 capsules.

Blisters of 10 capsule. Boxes of 10, 20 and 30 capsules.

STORAGE:

Store in a dry place below 30°C.

Protect from light.

Keep out of reach of children.

Jauhkan dari kanak-kanak.

MANUFACTURED BY:

Noripharma Sdn.Bhd. (792633-A)

Lot 5030, Jalan Teratai,

5 1/2 Mile off Jalan Meru,

41050 Klang,

Selangor Darul Ehsan, Malaysia.

DATE OF REVISION:

December 2020